



AI FOR HR MANAGEMENT

EMPLOYEE PERFORMANCE AND PRODUCTIVITY DATA

PRESENTED BY KOSSALY DIANA & TEMIROV MADIYAR

PROJECT OBJECTIVE

- Build an AI-powered HR Intelligence System
- Analyze employee productivity and workforce behavior
- Predict attrition and workforce risks
- Improve recruitment and employee performance management
- Support HR decision-making using Power BI and Machine Learning
- Ensure fairness, ethics, and compliance in HR analytics

EXECUTIVE HR COMMAND CENTER

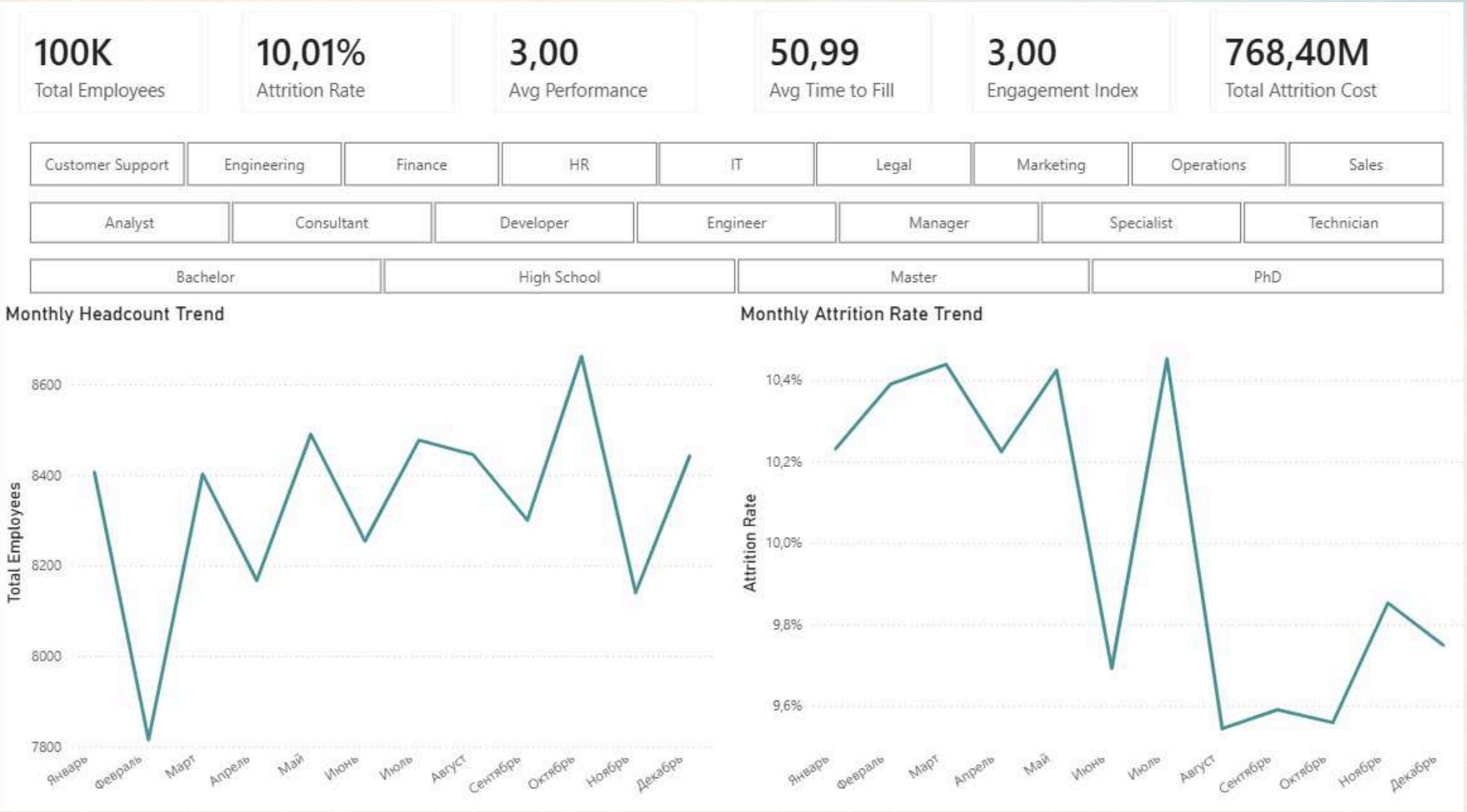
- Total Employees: 100K
- Attrition Rate: 10.01%
- Average Performance Score: 3.00
- Average Time to Fill: 50.99
- Engagement Index: 3.00
- Total Attrition Cost: 768.40M

Dashboard Insights:

- Monthly headcount remained relatively stable over time
- Attrition trends fluctuated across different periods
- The company faces high financial losses caused by employee turnover

Main Result:

- Attrition creates a major financial impact, costing the company over 768M.



RECRUITMENT INTELLIGENCE

- AI-ranked candidate pipeline analysis
- Hiring diversity analysis by gender
- Hiring velocity comparison by department
- Talent quality distribution:
 - Need Improvement
 - Solid Performer
 - Rising Star

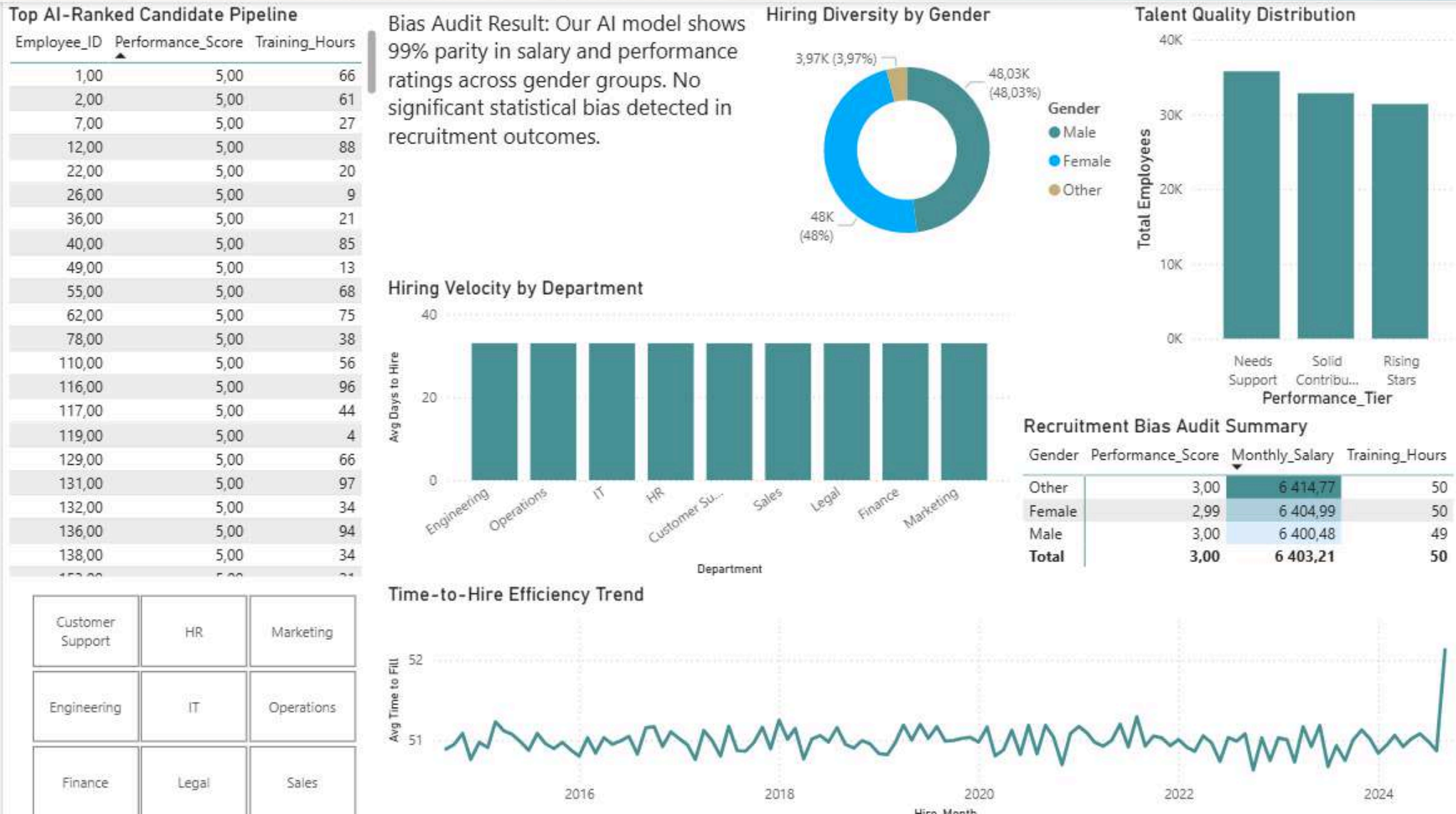
- Recruitment bias audit summary

Dashboard Insights:

- Male and female hiring distribution is relatively balanced
- No statistically significant recruitment bias detected
- Departments maintain similar hiring speed levels

Main Result:

- The recruitment model achieved fair and consistent hiring recommendations.



ATTRITION RISK CENTER

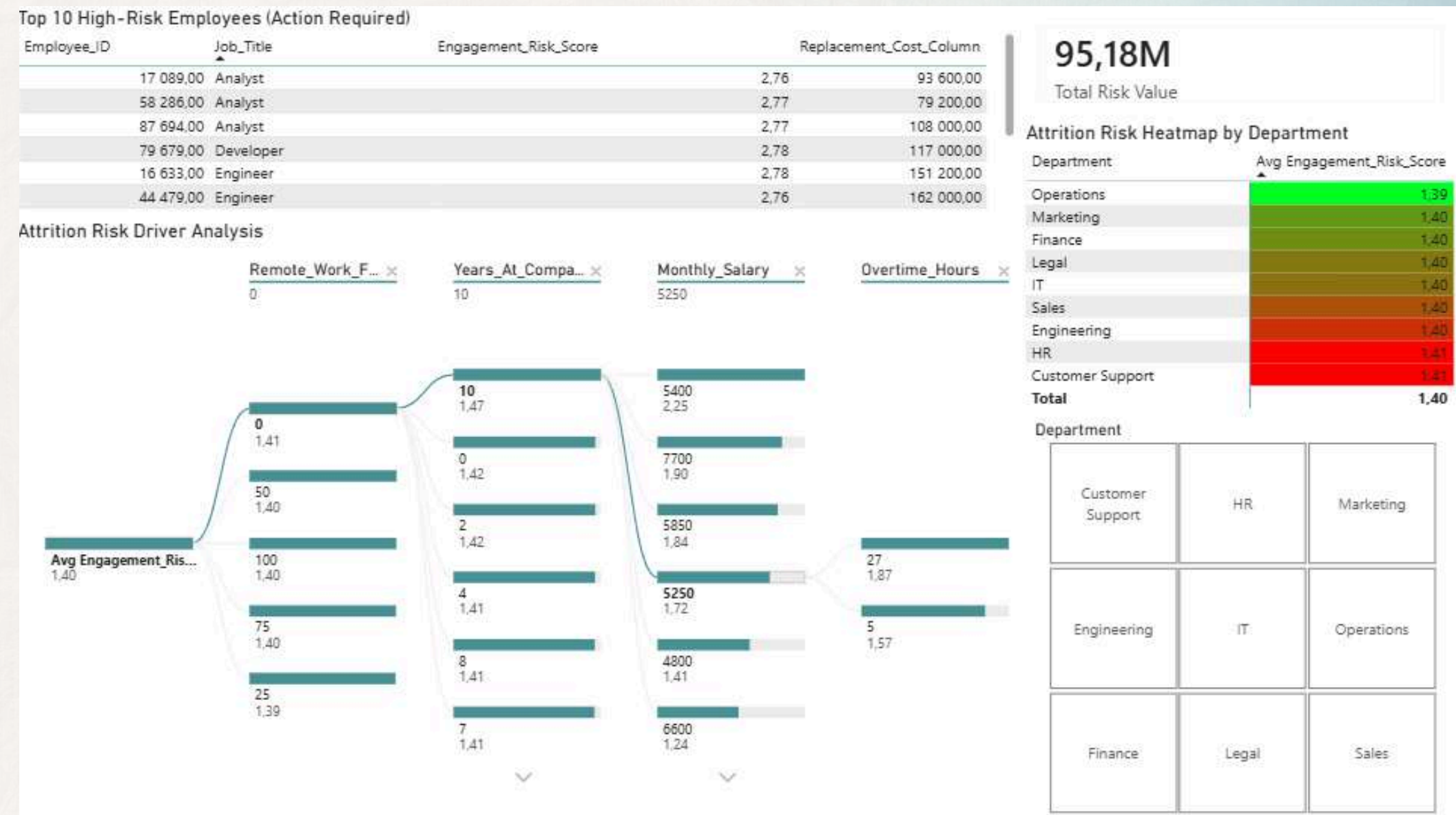
- Top 10 high-risk employees identified for HR intervention
- Attrition heatmap by department
- Attrition driver analysis based on:
 - Remote work ratio
 - Years at company
 - Monthly salary
 - Overtime hours
- Total Risk Value: 95.18M

Dashboard Insights:

- Sales and Operations showed higher attrition risk scores
- Employees with high overtime and lower engagement had higher resignation risk
- Salary and work-life balance strongly affected retention

Main Result:

- Overtime and low engagement were the strongest predictors of employee attrition.



PERFORMANCE INTELLIGENCE

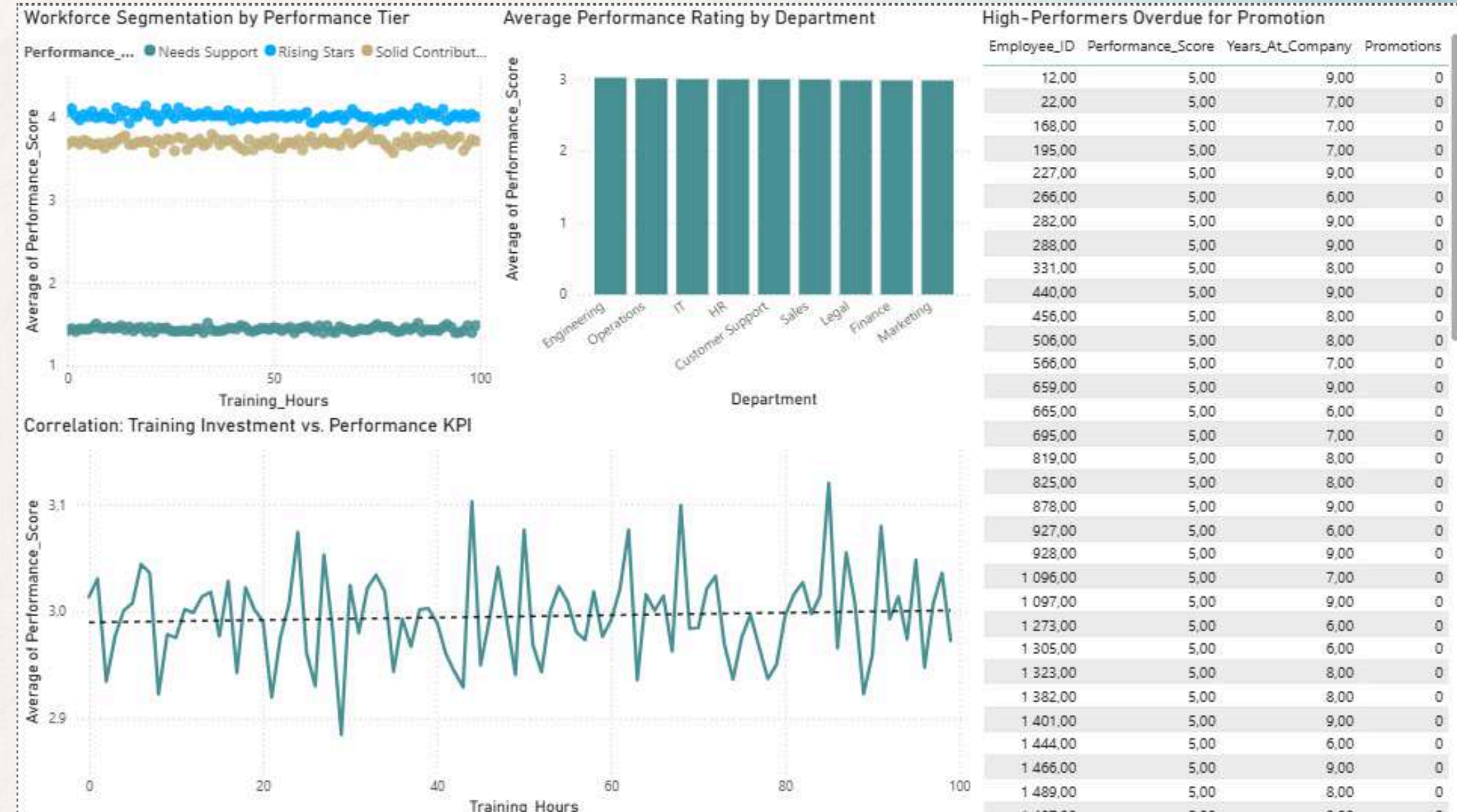
- Workforce segmentation into:
 - Needs Support
 - Solid Contributors
 - Rising Stars
- Average performance rating by department
- Training investment vs performance analysis
- High performers overdue for promotion identified

Dashboard Insights:

- Most departments maintained similar average performance levels
- Employees with more training hours generally achieved stronger KPIs
- Several high-performing employees had not received promotions despite long tenure

Main Result:

- Training and promotion opportunities directly influence employee productivity.



ENGAGEMENT & WELLBEING MONITOR

- Employee satisfaction analysis by department
- Teams requiring urgent HR intervention highlighted
- Overtime burden used as stress indicator
- Absenteeism trends by department
- Wellbeing Risk Index: 71.97

Dashboard Insights:

- Engineering and Operations showed higher wellbeing risk levels
- Overtime remained consistently high across departments
- Average satisfaction score stayed around 3.00

Main Result:

- High overtime levels increased wellbeing risks and absenteeism.



WORKFORCE PLANNING & FORECASTING

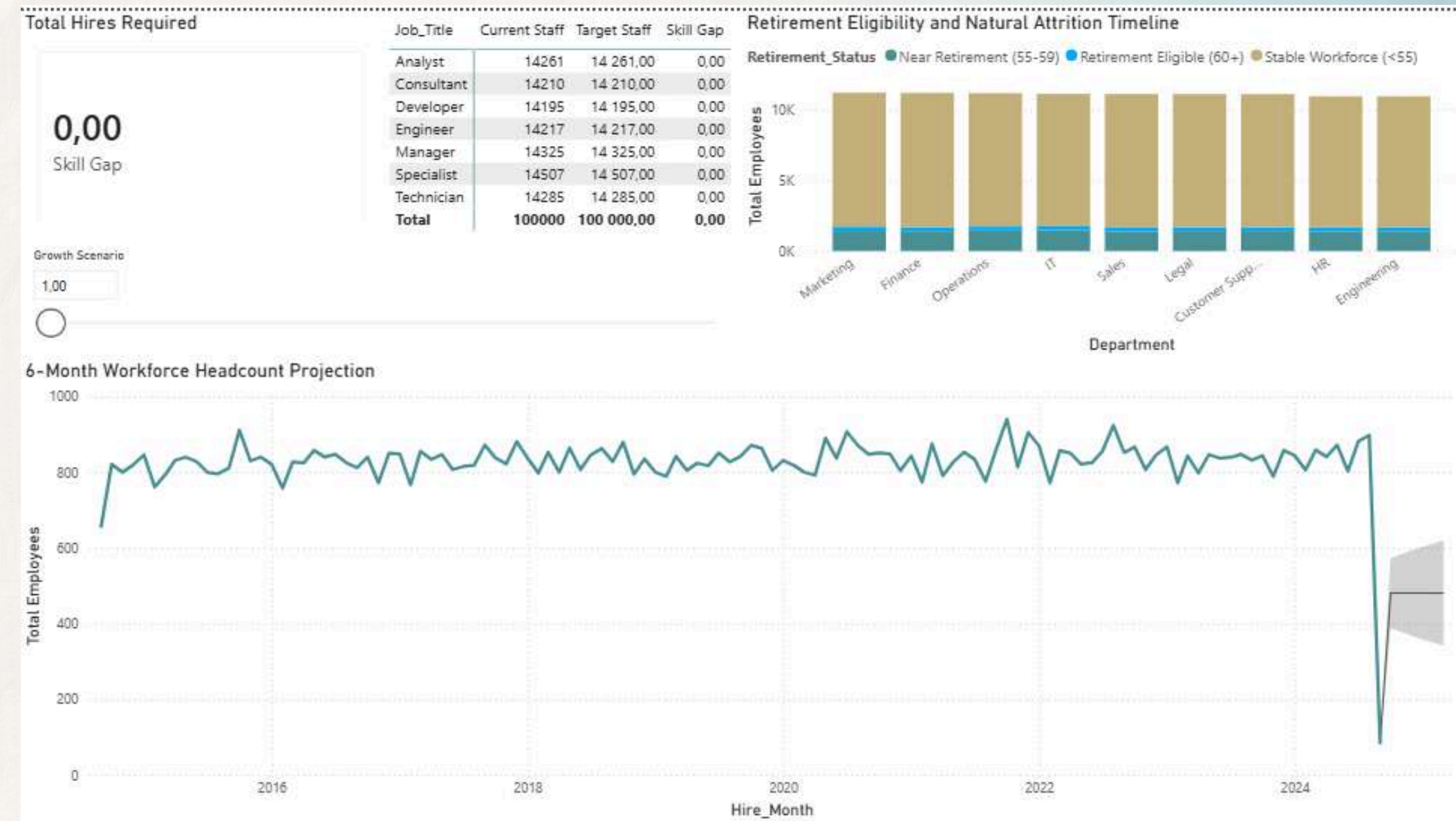
- 6-month workforce headcount projection
- Retirement eligibility and natural attrition analysis
- Skill gap monitoring dashboard
- Workforce planning by job title

Dashboard Insights:

- Workforce levels remained relatively stable over time
- Some job roles may require future replacement planning due to retirement trends
- Current skill gap value remained close to zero

Main Result:

- Forecasting helps HR prepare future hiring and workforce replacement strategies early.



ETHICS, BIAS & COMPLIANCE

- Bias audit across:
 - Gender
 - Age groups
- GDPR compliance scorecard:
 - Data Minimization — PASS
 - Purpose Limitation — PASS
 - Transparency — PASS
 - Right to Explanation — PASS
 - Ethics Risk Register — PASS

- AI transparency report included

Dashboard Insights:

- Attrition prediction rates stayed close across all demographic groups
- No major discrimination patterns detected
- SHAP values used to explain AI decisions transparently

Main Result:

- The AI system demonstrated fair prediction behavior and strong compliance standards.

Bias Audit: Risk Prediction Rates by Protected Groups

Gender	Avg Engagement_Risk_Score	Attrition Rate
☐ Female	1,40	10,03%
20-29	1,40	10,00%
30-39	1,40	9,97%
40-49	1,39	10,14%
50-59	1,41	10,09%
60+	1,38	9,32%
☐ Male	1,40	10,01%
20-29	1,41	9,60%
30-39	1,39	9,98%
40-49	1,40	10,09%
50-59	1,41	10,12%
60+	1,39	11,65%
☐ Other	1,40	9,75%
20-29	1,37	10,92%
30-39	1,41	9,15%
40-49	1,41	9,90%
50-59	1,40	8,76%
60+	1,41	14,16%
Total	1,40	10,01%

GDPR Compliance Scorecard

Principle	Status	Mitigation Strategy
Purpose Limitation	PASS	Data used strictly for HR analytics and retention.
Data Minimization	PASS	Only necessary workforce metrics are processed.
Right to Explanation	PASS	Plain-language reports generated for each model.
Transparency	PASS	SHAP values explain individual risk factors.
Right to Erasure	PASS	System supports automated data deletion requests.

AI Ethics Risk Register

Ethical Concern	Severity	Mitigation Implemented
Algorithmic Bias	High	Monthly bias audits by gender and age group.
Automation Bias	Low	Final decisions always require HR manager review.
Data Privacy	Medium	End-to-end encryption of salary & personal data.

Model Transparency Report

Our Attrition Model (XGBoost) makes decisions based on objective data: salary competitiveness, overtime hours, and tenure. It does not use protected attributes for prediction. We use SHAP (Shapley Values) to ensure every risk score can be explained to the employee.

HR TRANSFORMATION ROADMAP

Potential Annual Savings:

- 350K

90-Day Implementation Roadmap:

- Quick Wins Phase
- Integration Phase
- Full Roll-out Phase

Measurable Business Value:

- 30% reduction in voluntary attrition
- 50% faster hiring for critical roles
- Improved GDPR and ethics compliance
- Better succession planning
- Improved employee wellbeing monitoring

Main Recommendations:

- Launch retention programs in high-risk departments
- Increase promotion opportunities for top performers
- Reduce overtime in Engineering teams
- Continue quarterly AI bias audits





THANK YOU